

Skeptical Yet Curious: Attitudes Towards Conversational Assistants for Supporting Social Engagement among Older Adults Living Alone

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Abstract

As people age, forming and maintaining social connections can become increasingly complex, especially for those living alone. Conversational assistants (CAs) show promise for supporting older adults' socio-emotional well-being; yet, prior work has primarily focused on CAs potential as digital assistants or companions rather than tools for fostering real-world human connections. We aim to bridge this gap by exploring how CAs can facilitate meaningful social interactions. Through nine scenario-based interviews with older adults living alone, we examined their perceptions of CAs that support relationship-building and mediate meaningful connections, focusing on connecting with others rather than offering companionship. Participants were skeptical yet curious about CAs for socialization. They showed interest in the concepts due to their potential for accessible and personalized socio-emotional connections. However, they also raised significant concerns related to privacy and relational trust. We offer insights into how to better design CAs to meet the socio-emotional needs of older adults.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; • **Social and professional topics** → **Seniors**.

Keywords

Aging, Conversational assistants (CAs), Older adults, Living alone, Social support

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1 Introduction

As the global population ages, many older adults are increasingly living alone due to factors such as longer life expectancy, loss of a partner, or a desire for independence. Within the field of Human-Computer Interaction (HCI), technologies like conversational assistants (CAs) have been explored as tools to support older adults' emotional and social needs. Existing systems such as ElliQ and Billie have demonstrated the potential of CAs to act as digital companions by simulating conversation and suggesting activities to combat loneliness [6, 17]. However, most research has focused on using these technologies as substitutes for human connection rather than as facilitators of real-world social interactions. This approach overlooks the diverse social needs of older adults living alone, particularly some older adults' desire for meaningful connections with others who share similar experiences or challenges [21]. By shifting the focus from companionship to social facilitation, this study explores older adults' perceptions of how Conversational Assistants (CAs) could mediate interactions and build trust in two specific contexts: (1) forming new connections with peers for caregiving support and (2) maintaining existing relationships through accessible communication technologies.

We employed scenario-based interviews with nine older adults aged 65 and above, all living alone, to explore their perceptions of conversational assistants (CAs) that support their social engagement with other individuals in the two storyboard contexts. Participants provided feedback to help us identify their needs, concerns, and attitudes toward the potential use of CAs in enhancing social interactions with other humans. The findings revealed that

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participants viewed conversational assistants as potentially valuable tools for managing connections with others, particularly for socio-emotional support and accessibility. However, participants expressed skepticism about the assistants' ability to address trust concerns, especially when connecting with strangers. This research contributes to HCI by exploring the attitudes of older adults living alone toward CAs that support connections with other individuals, particularly for socio-emotion support and improved accessibility. We also contribute design opportunities for addressing the unique needs in the future by emphasizing the importance of personalized recommendations, gradual relationship-building mechanisms, and trust-enhancing features in CA design.

2 Background and Related Work

Several factors, including increased life expectancy, loss of a partner, need for independence, or financial means, can contribute to older adults living alone [21]. Living alone can offer benefits for older adults, such as exposure to diverse social networks [10], greater autonomy, and reduced stress [11]. However, it can also bring significant challenges. For many older adults, this independence often comes with increased risk to physical, social, emotional, and financial well-being [21]. Therefore, within HCI and Health literature, researchers have focused specifically on studying the role of technology in supporting the health and well-being of older adults living alone [15, 32].

Conversational assistants, including voice assistants and chatbots, are considered beneficial for older adults due to their potential to mitigate accessibility challenges [26, 30]. However, due to the use of a conversational approach that mimics human conversation, conversational assistants are also perceived as potentially beneficial for simulating conversation and engaging in social interactions with older adults as digital companions [6, 17, 27]. Researchers have studied the ways conversational assistants can support older adults living in different environments, finding interest and use for social support in both community-dwelling [24, 27, 29, 31], and community-living environments [14, 25]. Others have developed assistants to support older adults socially as digital companions [8, 16, 24]. For example, ElliQ proactively engages seniors in conversation and suggests activities to combat loneliness [6]. Another group of researchers developed a system with a virtual assistant, "Billie," that focused on providing an easy-to-understand-socially cooperative dialogue exchange with older adults to have empathy towards older adults as a companion rather than as an assistant [17]. More recently, researchers are exploring how to use conversational assistants to support the unique needs of older adults who live alone. For example, researchers have studied how lower-income older adults living alone use commercial assistants to support everyday and emotional tasks, finding that older adults found the assistant useful for supporting everyday tasks and providing some emotional support as a conversational partner and engaging in enjoyable activities (i.e., listening to music) [15]. Collectively, these innovations significantly enhance the independence and quality of life for seniors living alone. Yet, most research on conversational agents (CAs) for older adults focuses on using them as digital companions to reduce loneliness and provide emotional support. Systems like ElliQ [6] and Billie [17] simulate companionship by

conversing with older adults and suggesting activities. While these technologies are helpful for emotional well-being, we are still learning how to design these types of CAs to support older adults with different social needs and experiences, including needs for real-world social connections.

Our research aims to fill this gap and extend prior work by examining how CAs can help older adults living alone connect with others for social purposes. We explore older adults' perceptions of CAs for mediating human-human communication to actively support social interactions and strengthen relationships instead of serving as a digital companion. We aim to provide new insights into how CAs can improve social engagement and emotional support with others to enhance the independence and quality of life for seniors living alone.

3 Scenario-Based Interviews

We conducted nine scenario-based user interviews to help participants visualize potential future interactions with conversational voice assistants that mediate social interactions. Scenario-based design practices are well-established user-centered design approaches used in Human-Computer Interaction (HCI) and are commonly used early in design to vet early design ideas [28]. The approach is an alternative to the solution-first approach, which emphasizes the rapid implementation of design solutions to gather feedback and instead emphasizes potential users and their needs, user reflection, and flexible solutions that can be easily changed [28]. We chose this method because it aligned with our research goal. Our goal was to obtain feedback on early ideas but also to identify the needs and pain points of older adults regarding their socialization, technology use, and the potential for voice assistants. Our study was reviewed and approved by the Institutional Review Board (IRB) at our university before we recruited participants or collected any data.

3.1 Participants and Recruitment

For recruitment, we distributed flyers and contacted individuals working within older adult communities. We also distributed our study information through a health study recruitment listserv to reach a more diverse audience. To screen participants once they expressed interest, we followed up with a phone call to confirm that they met our study criteria and to schedule an interview. We recruited nine participants who were (1) aged 65 or above, (2) interested in new social wellness technology, and (3) living alone.

Most participants were in the age range of 65–69 (N=3) or 70–74 (N=4), indicating younger, older adults (Table 1). Most participants were female (N=7). All participants lived alone and self-reported it was due to divorce, loss of a spouse, or never being married; however, some mentioned other factors, including relocation or difficult family situations. Most participants (N=6) used technology frequently to support social activities, and all others (N=3) used technology occasionally.

4 Interview Procedures

We began the interview by providing participants with an overview of the study, followed by questions to understand their current social experiences and challenges. Interviews included exploring their social engagement and strategies, as well as the barriers they

Table 1: Participant Demographics and Technology Use

P No.	Age Range	Gender	Marital Status	Ethnicity	Technology Use (Social)
P1	65-69	F	Widowed	White/Caucasian	Occasionally
P2	65-69	F	Single	Black/African American	Frequently
P3	65-69	M	Divorced	Hispanic/Latino	Occasionally
P4	70-74	F	Divorced	Black/African American	Frequently
P5	70-74	F	Single	White/Caucasian	Frequently
P6	70-74	M	Widowed	Hispanic/Latino	Frequently
P7	70-74	F	Divorced	White/Caucasian	Frequently
P8	75-79	F	Widowed	White/Caucasian	Frequently
P9	80-84	F	Widowed	White Caucasian	Occasionally

face. We also asked participants about their use of technology for social wellness. Participants were presented with two storyboards and asked for feedback on the storyboards, including their initial thoughts on the concept and the use of conversational assistants, as well as any concerns and suggested changes.

We developed the storyboards with the goal of understanding participants' perceptions about the proposed use of conversational assistants in two specific contexts: supporting connections with other individuals they may not know and supporting connections with individuals they already know. These two scenarios aligned with the National Institute of Health's recommendations of building and maintaining new connections as part of continued social wellness [23] and had some overlap. Our goal was to assess their general openness to adopting conversational assistants and enhance social interactions in the specific contexts we presented.

The first storyboard (Fig. 1) depicts the experience of Lucy, an older adult caregiver struggling with managing her household while caring for her husband, who is chronically ill. She installs an app called "We Care" that includes a voice assistant named Eva. Eva offers emotional support and caregiving tips but can also connect with other caregivers experiencing similar challenges. Based on recent interactions with Lucy, Eva recommends that Lucy connect with Mary, another caregiver, to build her support network.

We chose the first storyboard narrative to showcase how conversational technology might provide personalized support and facilitate peer connections while also addressing caregiver burnout and isolation, another risk to social wellness. The scenario highlights the potential of conversational apps to support caregiver well-being through practical advice and building new connections. Finding new connections is a key aspect of the narrative to emphasize the importance of peer support in alleviating the emotional burden of caregiving. By connecting Lucy with Mary, the app suggests the potential for conversational technology to assist in creating a supportive network for caregivers to also improve emotional wellness.

In the second scenario (Fig. 1), an older adult, Emma, receives a text message from her daughter, Sonia, inviting her to an anniversary celebration and asking for cake suggestions. Due to dexterity challenges and inconsistent auto-correct, Emma finds it difficult to type and struggles to reply. Therefore, Emma uses a voice assistant, which reads the message aloud to her and allows her to respond verbally to the message. Emma replies verbally to the text, confirming her attendance and suggesting a cake, with the voice

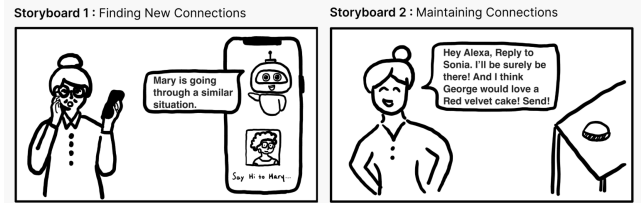


Figure 1: Examples of storyboard frames we designed depicting 'Finding New Connections' and 'Maintaining Connections'.

assistant sending the message on her behalf. We chose this storyboard to illustrate how voice-assisted technology can empower individuals with accessibility needs to maintain communication and social connections. The narrative presented is commonly available in existing voice assistant technologies such as Siri, so we wanted to explore if there was potential to enhance this. This narrative also highlights the significance of maintaining new connections, as the voice assistant enables Emma to maintain interaction with her daughter through text. By addressing accessibility barriers, the story underscores the potential of such technology in enhancing the quality of life for individuals with accessibility needs.

All interviews were audio recorded and transcribed with participant permission. To analyze the interview data, we first transcribed and cleaned the transcripts. Later, using a reflexive thematic analysis [2], we developed low-level codes inductively that were related to recurring participant comments on storyboards. We grouped codes into higher-level themes through iterative discussion sessions and affinity mapping of lower-level codes into broader categories. To quantify participants' initial perspectives of the two concepts, we categorized their responses to questions about the helpfulness of the approach based on their direct responses (i.e., they responded that the concept would be helpful for them or not). Final themes were related to initial thoughts about and potential future use of the CAs presented in the storyboard by the participant or older adults', more generally, concerns and areas for improvement.

5 Findings

Findings revealed that participants viewed conversational assistants such as chatbots and voice assistants as potentially valuable tools for

helping them manage their social connections with others. However, some participants shared their skepticism about the ability of these types of systems to address their trust concerns.

5.1 Helpful for Connecting for Socio-Emotional Support, Family, and Friends Not So Much

Of the nine participants, four felt an assistant that connects them to others for caregiving support would be helpful for older adults, and five shared that it could be helpful to them in the past or in the future. For example, one participant, P3, shared, *"I think a technology platform such as this would be helpful. I would have liked to have something like that when my mom was living with me and such, because there's tons of adults that are taking care of their elderly parents and such, they would have been very helpful on that. I like the concept."* Of those participants who saw value, they mostly commented on the system's ability to coordinate and connect them with other people who were going through similar things. One participant, P9, reflecting on how others seeking emotional support might use this technology, added, *"If I'm unhappy about anything, I go take a walk. I go do something, but I can definitely see where people might get [socio-emotional] relief or understanding with this kind of technology."*

Fewer participants expressed value in using conversational assistants to connect with family and friends. Three participants said that they see conversational assistants as a useful way to directly communicate with other individuals they know – such as friends and family. Several noted that the scenario seemed familiar to their current use of voice technologies to connect with others. For example, one participant, P1, acknowledged, *"I kind of use something like this in my car. Oh, you know, so you can push the button and say, send a text, whoever. And, yeah, I like that feature."* However, they also acknowledged they may like to see additional features for connecting in the future, *"I would love to see, like emergency numbers or something. Being able say, you know, help I've fallen, and I can't get up call 911, you know"*. Another shared that they might consider purchasing a commercial voice assistant to support this sort of communication in the future. P2 shared, *"I'm thinking that I'm going to start using some of these assistants, because they seem to be very helpful ... Since I've been introduced to how they work and what they can help assist me with so I may pick up one of them."*

5.2 Providing Accessibility and Facilitating Personalized Connections

Participants felt that the primary value of using conversational assistants to connect with others directly, without any additional features, was related to providing accessible connections through multimodal features. Yet, the findings revealed a duality among participants, noting the potential of CAs to enhance accessibility while raising concerns about dependence on technology. Three participants shared that they would consider conversational assistants for accessibility reasons only. Yet, all participants recognized the value of conversational assistants for improving the day-to-day accessibility needs of older adults, particularly for supporting individuals with physical, cognitive, or visual impairments in connecting with others socially. For instance, P2, referring to storyboard 2,

emphasized their potential utility for those with arthritis or learning disabilities, *"The assistant [could] really assist them [other older adults] to talk it, when they really need to message the person that they're talking to. So, I think that's one of the better things about the assistant that I'm impressed with."* Similarly, P6 acknowledged the assistants' potential for supporting individuals with visual impairments, *"somebody who is visually impaired, something like that might be very helpful."* Sharing their personal experience, P9 expressed how the accessibility features of conversational assistants have been particularly beneficial for them socially: *"I'm learning to use voice to text because I have an issue with my left hand when writing, and I'm lefty."* However, other participants also shared contrasting views, noting that they likely would not adopt CAs that support social interactions for accessibility reasons only because they personally did not have current accessibility needs. P6 stated, *"I don't think I would use it [conversational assistant] otherwise right now for that [connecting with people directly] because I'm still able to see and read text messages."* P8 similarly shared, *"I am healthy; I like to be independent and not depend on technology."* These participants felt that they currently did not have accessibility needs and, therefore, would not adopt a conversational assistant for social purposes for this reason alone, highlighting a dichotomy in perspectives.

5.3 Skeptical about Matchmaking and Building Relational Trust

While participants felt conversational assistants could address accessibility challenges related to connecting socially, most participants saw value in the potential ability of conversational assistants to help support them in finding relevant connections, as evidenced by their high interest in the caregiving scenario. Yet, despite their enthusiasm, participants raised concerns about trusting an automated matchmaking system that connects individuals. The findings revealed complex attitudes toward emotional disclosure with other individuals through conversational assistants, highlighting opportunities and hesitations in using this type of technology for facilitating individual social connections.

Participants expressed a notable dichotomy between an AI-enabled interface alleviating some of the burden of finding connections and their apprehensions and comfort level in connecting with unknown individuals. P1 shared their concerns, *"I don't know if I would use that or not. Talking to a stranger about my problems, I don't know, not sure."* Other participants noted that even if they opted to use the technology, it would take time to build trust; P6 shared, *"I wouldn't feel comfortable with it [the conversational assistant] right away. It would take some time for me to get used to working with something like that. And again, like I said, it's a small matter of trust here, on this basis, connect me with somebody else, someone I don't know?"* While participants saw value, they noted relational concerns they would like addressed before adopting such technology. Therefore, participants suggested implementing measures like user vetting to ensure credibility; as P4 proposed, *"these people would be vetted when they're coming on the platform."*

6 Discussion

Overall, the findings highlight both opportunities and challenges for voice assistant adoption. While they can empower individuals

by enhancing accessibility and supporting autonomy in daily activities, there is also a need to address concerns about over-reliance on technology and its potential impact on relationship building and personal independence. While there have been numerous studies exploring older adults use of CA technologies [16, 26, 27, 30, 31], to support health-related tasks [4, 5, 7, 12, 19, 22] including caregiving support [13, 33, 34], these studies often focus on the daily management of physical health tasks or activities of daily living compared to a dedicated focus on social wellness [23] and do not emphasize ways to support human to human connections for socio-emotional health. While other studies have established the benefits of CAs for supporting older adults' social health, specifically [6, 17, 24, 25, 27, 29, 31], they do not explore how CAs might mediate human to human connections and mainly focus on CAs as digital companions [8, 16, 24]. Yet, prior work suggests that older adults see benefits in exploring CAs that can support social wellness goals with others [14, 30]. Based on the findings from the study, we provide design considerations for conversational assistants that support older adults living alone.

6.1 Moving Beyond a Tool for Direct Communications, Conversational Assistants to Facilitate Connection

Findings revealed that participants believed conversational agents were a promising opportunity for providing accessible management of social connections with others. However, they also suggested they would value **CAs that go beyond direct communication**. Other researchers have found that older adults value CAs, in particular voice assistants, as the potential for addressing accessibility concerns [26] and that older adults use existing voice assistants for social purposes [25, 30, 31]. Our findings build on this past work, suggesting that participants also saw CAs as a potentially accessible option for connecting with others socially. Yet when presented with the option of CAs that facilitate direct communication, some participants noted that they would likely not use the feature unless they had an accessibility concern because they would prefer to call directly or that they already used voice technologies in this way. Instead, participants preferred the personalization offered in the second scenario. Based on analyzing user preferences and behaviors, the CA could provide personalized recommendations for connecting with other individuals with similar experiences or needs. Other researchers have found older adults with visual disabilities see voice technologies such as interactive voice response (IVR) [3] and voice assistants [29] as useful for improving access and allowing them more freedom to participate in social engagements. Our findings suggest that when designing CAs for individual older adults' social support, we should also **consider how CAs might provide personalized recommendations that align individuals with similar interests**. In particular, our findings suggest older adults see CAs potentially acting as intermediaries, helping to coordinate social interactions by managing schedules, sending reminders, or even facilitating introductions between like-minded individuals.

6.2 Supporting Initial Engagement and Relational Trust

Findings suggest opportunities for CAs that can facilitate connections among older adults with aligned interests, particularly when seeking or needing socio-emotional support. However, we suggest that **CAs designed for social support should have the ability to learn and adjust suggestions over time to ensure that the social facilitation remains relevant and engaging** as the older adult's needs and preferences evolve. In addition, we found that in the context of supporting socio-emotional connections, it is important for older adults to be able to control the information disclosed about them and how the relationship engagement unfolds. Prior work notes that individuals [1, 18, 20], including older adults, have expressed privacy and confidentiality concerns when engaging with some CAs, such as voice assistants, due to the limited transparency provided regarding how data is collected, stored, and used. However, our participants noted a different type of privacy concern related to personal privacy and relational trust. While they appreciated having personalized help to support initial engagement, they were skeptical about the systems' ability to connect them to other relevant individuals they would be comfortable opening up to. Their attitudes were partly due to trust concerns about connecting safely to other people but also related to the need to build a trusting relationship once connected. We found that voice assistant platforms that facilitate social interactions, particularly when matching users with others who have similar experiences, **should incorporate features that enable controlled, gradual relationship building while paying special attention to individual comfort levels**. Our interviews suggest that designers must also consider that relationships often require reciprocity and trust building; therefore, **in addition to proper screening procedures, CAs that facilitate these types of matches must consider ways to ethically support this trust and reciprocity among individuals, including adding proper vetting mechanisms**.

7 Limitations

It is crucial to acknowledge the shortcomings of our study. We used a scenario-based method to help participants imagine future uses of CA technologies that support their social wellness needs. While this approach is well-established as a valid and useful methodology in Human-Computer Interaction [28] to support user-centered and participatory design processes to vet early ideas, it has limitations. In particular, while scenario-based design can be advantageous for imagining real-world use, the success of the process depends on those involved in the evaluation. Aligned with the goals of qualitative inquiry [9], our goal is not to generalize, however we acknowledge that opinions may vary among older adult groups with different characteristics. Further, while the storyboards provided context through examples of real-life scenarios, they may not fully capture how trust would evolve in real-world interactions with a working prototype. Future studies could implement interactive prototypes for different use cases to better understand how older adults' perceptions of trust change over time. However, we believe our findings may transfer to other groups of older adults with similar characteristics.

8 Conclusions

In the paper, we explored perceptions of conversational assistants (CAs) for supporting connecting with others among older adults living alone. Through nine scenario-based interviews, we presented two concepts for CAs that connect older adults to other individuals as opposed to a CA that serves as a digital companion. Overall, participants appreciated CAs in general due to their ability to provide multi-modal interactions addressing potential accessibility needs. They liked the idea of CAs that could provide personalized support for finding relevant connections for socio-emotional support. However, participants remained skeptical about CA's ability to find relevant matches and expressed hesitation about potentially opening up to someone they do not know. Based on these findings, we discuss design opportunities for CAs that help connect older adults living alone with other individuals. In the future, we will expand the study to hear broader perspectives from older adults who might benefit from CAs as social technologies.

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